UTKARSH ARYAN MISHRA

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SUMMARY

Final-year B.Tech IT student targeting Software Engineer roles, with two internships and hands-on backend and cloud experience. Built a 25+ endpoint Node.js/Express/MongoDB backend for UPPCL, deployed on AWS (Terraform, Docker, Nginx, GitHub Actions), and an event-driven deployment platform in TypeScript. AWS Certified Cloud Practitioner; 100+ LeetCode problems solved; proficient in AI-assisted development (Antigravity, Claude, ChatGPT, Codex).

TECHNICAL SKILLS

Backend Development: Node.js, Express.js, TypeScript, REST APIs, Microservices, MongoDB, PostgreSQL, MySQL, Redis
Programming: Python, JavaScript, TypeScript, Java, C, React
Core CS Fundamentals: Data Structures & Algorithms, OOP, System Design, DBMS, Operating Systems, Computer Networks
Cloud & Infrastructure: AWS, EC2, EKS, S3, CloudFront, IAM, Terraform, Linux
Containers & Orchestration: Docker, Kubernetes, Helm, NGINX Ingress
DevOps & GitOps: GitHub Actions, Argo CD, CI/CD, Infrastructure as Code, Git, GitHub
Observability & Security: Prometheus, Grafana, Trivy, Kubernetes HPA, NetworkPolicies
Testing & Tools: Unit Testing, Jest, Supertest, Postman
AI & Developer Tooling: Antigravity, Claude, ChatGPT, OpenAI Codex, AI-Assisted Development, Prompt Engineering

EXPERIENCE

Software Engineer Intern | UPPCL — Uttar Pradesh Power Corporation Ltd |. [Certificate] Jun 2026 – Jul 2026

IT Division, Lucknow | Backend Development · Cloud Deployment · Docker · Terraform · Nginx

- Built the backend for the Daily Revenue Monitoring Portal (DRMP), aggregating revenue data from 3 payment streams (Bank Wire, Payment Gateway, Billing System) across UPPCL divisions; designed 25+ REST API endpoints with JWT auth, bcrypt, RBAC, and server-side validation.
- Designed 4 MongoDB collections (Users, Payment Submissions, Consolidated Reports, Audit Logs) via Mongoose ODM with an asynchronous audit-trail logging service, prior-date continuity checks, and PDFKit/ExcelJS report exports.
- Tested 25+ REST endpoints with Jest, Supertest, and Postman. Portal was hosted briefly on UPPCL servers, then redeployed on AWS EC2 via Terraform with Docker (non-root, multi-stage), Nginx, Certbot SSL, and GitHub Actions CI/CD; used Antigravity to draft and debug configs.

Software Engineer Intern | i2k2 Networks Pvt. Ltd. | [Certificate] Jan 2026 – Apr 2026

Noida, Uttar Pradesh | Cloud & Managed Services Company

- Deployed internal applications and built GitHub Actions CI/CD pipelines while working alongside an engineering team.
- Used Git/GitHub branching workflows, Docker, and AWS to learn how production SDLC and deployment practices work in industry.

PROJECTS

Kubernetes Platform Engineering & GitOps Lab | Repo

Kubernetes · Helm · Argo CD · Terraform · Prometheus · Grafana · Trivy

- Architected modular Terraform IaC provisioning a multi-AZ AWS VPC (4 subnets, 2 AZs) and EKS v1.30 cluster, with Argo CD + Helm v3 automating GitOps delivery — achieving commit-to-production rollout in under 3 minutes with zero manual cluster access.
- Engineered hardened multi-stage Docker builds gated by Aqua Trivy CVE scanning, cutting image size 79% (1.1GB → 227MB) and blocking High/Critical vulnerabilities from reaching the registry.
- Configured Kubernetes HPA (2→5 replicas at 60% CPU) and Pod Disruption Budgets (minAvailable: 2); chaos testing verified sub-2-second failover with zero dropped requests during simulated pod crashes.
- Instrumented RED-method observability (rate, errors, duration) via Prometheus Operator ServiceMonitors at 15-second scrape intervals, with latency histograms across 10 buckets (5ms–5s) visualized on Grafana dashboards.

DeployForge | Cloud-Native Internal Developer Platform & Deployment Engine | Repo

TypeScript, Node.js, BullMQ, Redis, PostgreSQL, Prisma, Docker, Trivy, AWS SDK (ECR/ECS/ALB), React

- Built an event-driven Internal Developer Platform decoupling deployment requests from background execution via BullMQ and Redis, supporting concurrent builds with isolated filesystem sandboxing.
- Developed a 10-state finite state machine orchestrating the microservice deployment lifecycle — Git checkout, multi-stage Docker builds, security scanning, and Blue/Green promotion — with a real-time dashboard streaming build logs.
- Implemented a DevSecOps gate using Trivy CLI to scan images for CVEs and secrets, halting the pipeline on any critical finding.
- Integrated AWS SDK v3 (ECR, ECS, ALB) to automate image publishing with SHA-256 digest auditing and Blue/Green promotion via ALB listener re-weighting.
- Built with AI-assisted workflows: Antigravity for development/debugging, ChatGPT for error diagnosis, Claude for docs; reviewed and tested all AI-generated code.

EDUCATION

Bachelor of Technology in Information Technology Sept 2023 – May 2027

Dr. A.P.J. Abdul Kalam Technical University | CGPA: 7.66

CERTIFICATIONS

AWS Certified Cloud Practitioner (CLF-C02) — Amazon Web Services | Issued Jul 2026 [Certificate]

ACHIEVEMENTS & LEADERSHIP

- Core Team Member, Google Developer Groups (GDG) — Rajkiya Engineering College, Mirzapur.
- Problem Solving — Solved 100+ LeetCode problems; also practice on Codeforces.
- Class Representative — Liaison between faculty and 60+ students; resolved academic and administrative concerns.
- Student Coordinator, Startup Spotlight (Skillancer Pvt. Ltd.) — Led event logistics and on-ground execution. [Certificate]